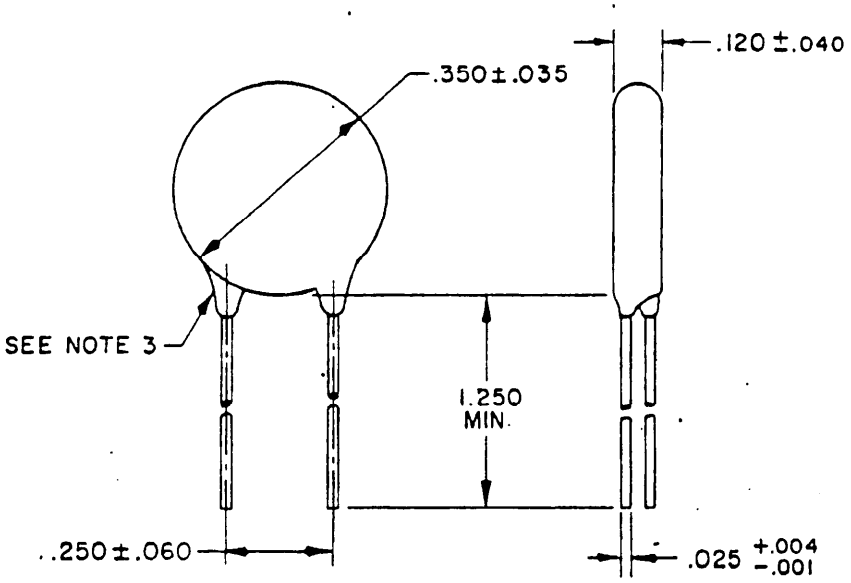
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), STYLE CK61

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-11015.



INCHES	MM
.001	.03
.004	.10
.025	.64
.035	.89
.040	1.02
.060	1.52
.120	3.05
.250	6.35
.350	8.89
1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Insulating coating shall not extend more than .125 (3.18 mm) along lead wires measured from a tangent to the coating surface drawn perpendicular to the lead wires.

FIGURE 1. Dimensions and configuration.

Ⓔ denotes changes.

FSC 5910

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TABLE I. Style CK61 characteristics.

Ⓔ Type designation 1/	Rated voltage	Rated temperature and voltage-temperature limits	Capacitance	Capacitance tolerance
	Volts, dc		pF	
CK61CZ2R2--	500	CZ	2.2	K, M
CK61CZ3R3--	500	CZ	3.3	K, M
CK61CZ4R7--	500	CZ	4.7	K, M
CK61CZ6R8--	500	CZ	6.8	K, M
CK61CZ100--	500	CZ	10	K, M
CK61CZ150--	500	CZ	15	K, M
CK61CZ220--	500	CZ	22	K, M
CK61CZ330--	500	CZ	33	K, M
CK61CZ470--	500	CZ	47	K, M
CK61CZ680--	500	CZ	68	K, M
CK61CZ101--	500	CZ	100	K, M
CK61CZ151--	500	CZ	150	K, M
CK61BX221--	500	BX	220	K, M
CK61CZ221--	500	CZ	220	K, M
CK61BX331--	500	BX	330	K, M
CK61CZ331--	500	CZ	330	K, M
CK61BX471--	500	BX	470	K, M
CK61CZ471--	500	CZ	470	K, M
CK61BX681--	500	BX	680	K, M
CK61AW222M-	500	AW	2,200	M

Ⓔ 1/ Where applicable, the complete type designation will include an additional symbol to indicate capacitance tolerance. The type designation will also include the letter "E" to indicate an epoxy coated capacitor (when applicable) or "-" will be deleted for wax impregnated case.

REQUIREMENTS

Design and construction:

Dimensions and configuration - See figure 1.

Ⓔ Case type - Disk, wax impregnated or epoxy coated (E).

Capacitance value - See table I.

Capacitance tolerance - ±10 percent (K) or ±20 percent (M) as shown in table I.

Rated temperature - -55° to +85°C (A); -55° to +125°C (B); or -55° to +150°C (C) as shown in table I.

Dielectric withstanding voltage:

Dielectric (DWV): Method 301 of MIL-STD-202.

Test voltage - 250 percent of rated voltage.

Body insulation: In accordance with MIL-C-11015.

Test potential - 1,300 volts dc.

Barometric pressure (reduced): Method 105 of MIL-STD-202. 0.82 inch of mercury (80,000 ft).

Test potential - 150 percent of rated voltage.

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Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 200,000 megohms, minimum.

Capacitance (Cap.): Method 305 of MIL-STD-202.

Test frequency:
100 pF or less - 1 MHz ±1 kHz.
Over 100 pF - 1 kHz ±100 Hz.

Test voltage - 1.0 ±0.2 volt rms.

Dissipation factor (DF): 1.5 percent, maximum (voltage-temperature limit X) or 2.0 percent, maximum (voltage-temperature limits W and Z).

Shock, medium impact: Method 205 of MIL-STD-202, condition C (50 G's).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 G's).

(c) Thermal shock and immersion:

Thermal shock: Method 107 of MIL-STD-202, condition A, except step 3 shall be at high test temperature.

Immersion: Method 104 of MIL-STD-202, condition B.

DWV - 250 percent of rated voltage.

IR - 150,000 megohms, minimum.

Salt spray (corrosion): Not applicable.

Terminal strength: In accordance with MIL-C-11015, 5 pounds applied.

Moisture resistance: Method 106 of MIL-STD-202.

DWV - 250 percent of rated voltage.

IR - 150,000 megohms, minimum.

Cap. - Within tolerance of table I value.

Solderability: Method 208 of MIL-STD-202. 2 terminals.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.

IR - 200,000 megohms, minimum.

ΔCap. - ±5 percent of initial measurement.

ΔDF - 0.5 percent, maximum.

Voltage-temperature limits: In accordance with MIL-C-11015.

Life (at elevated ambient temperature): Method 108 of MIL-STD-202. 2,000 hours (for qualification); 250 hours (performance check); 1,750 hours (continuation test).

Test potential - 200 percent of rated voltage.

DWV - 250 percent of rated voltage (at 25°C).

IR - 100,000 megohms, minimum (at high temperature and 25°C).

Cap. - Within tolerance of table I value (at 25°C).

DF - 1.5 percent, maximum (voltage-temperature limit X) or 2.0 percent, maximum (voltage-temperature limits W and Z) (at 25°C).

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Marking: In accordance with MIL-C-11015.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, MU
Navy - AS, OS, SH
Air Force - 11, 17
DLA - ES

User activities:
Navy - MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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